

6.Champagne Tower

Aim

To write a Python program using **Dynamic Programming** to determine the amount of champagne present in a particular glass of a champagne tower after pouring a given amount of champagne.

Procedure

1. Create a DP table to represent the champagne tower.
2. Pour the given amount of champagne into the top glass.
3. Store the amount in the top glass.
4. Traverse the tower row by row.
5. If a glass contains more than 1 cup, calculate the excess champagne.
6. Divide the excess equally between the two glasses below it.
7. Continue this process until the required row is reached.
8. Since a glass can contain at most 1 cup, use `min(1, amount)`.
9. Display the amount of champagne present in the requested glass.

main.py	Visualize	Run ▶	Output
<pre>1 def champagneTower(poured, query_row, query_glass): 2 tower = [[0.0] * (query_row + 2) for _ in range(query_row + 2)] 3 tower[0][0] = poured 4 for row in range(query_row): 5 for glass in range(row + 1): 6 if tower[row][glass] > 1: 7 extra = tower[row][glass] - 1 8 tower[row + 1][glass] += extra / 2 9 tower[row + 1][glass + 1] += extra / 2 10 tower[row][glass] = 1 11 return min(1, tower[query_row][query_glass]) 12 poured = 2 13 query_row = 1 14 query_glass = 1 15 print("Champagne in the glass:", champagneTower(16 poured, query_row, query_glass 17))</pre>			Champagne in the glass: 0.5

Result

Thus, the Python program successfully determines the amount of champagne present in the specified glass using **Dynamic Programming**.